



INTRODUCTION

Learn Malawi will become an innovative digital education platform designed to transform both primary and secondary education in Malawi by providing free, high-quality, and curriculum-aligned learning resources. Unlike conventional education websites, Learn Malawi will integrate adaptive learning technologies, offline accessibility, and interactive educational tools crafted specifically for Malawian students (Black & Wiliam, 1998). The platform will offer structured subject notes aligned with the Malawi national curriculum, PSLC, JCE and MSCE past papers with detailed marking schemes, interactive quizzes with instant feedback, multimedia tutorials such as videos, animations, and audio summaries, as well as teacher collaboration tools for resource sharing. This initiative aims to bridge the educational divide, particularly in rural and under-resourced schools and other public schools, where access to high-quality learning materials will remain limited (UNESCO, 2019).

BACKGROUND

Malawi's education system faces significant gaps, with urban schools benefiting from better access to textbooks, qualified teachers, and digital infrastructure, while rural schools, particularly primary schools in rural areas and Community Day Secondary Schools (CDSS), struggle with limited resources (MOE, 2023). The 2023 Malawi Education Sector Report highlights that only 42% of CDSS have adequate textbooks, forcing students to share outdated materials, while rural schools experience a 35% lower pass rate in national exams compared to urban schools. Additionally, fewer than 10% of schools have functional computer labs, limiting digital integration in classrooms. While some online learning platforms exist, they remain inaccessible to many students due to high data costs, lack of offline functionality, and content that does not align with the Malawi national curriculum. Learn Malawi will seek to bridge this gap by providing an offline-first design that enables resource downloads for Telecentres, AI-driven personalized learning with adaptive quizzes and progress tracking, and strict adherence to curriculum standards endorsed by the Malawi Institute of Education (MIE).

The platform will feature a responsive Progressive Web Application (PWA) architecture, incorporating adaptive content delivery, offline functionality, and USSD/SMS accessibility to ensure universal access across Malawi's technological landscape (Brown et al., 2014). Continuous monitoring of key performance indicators, including learning outcomes, platform engagement, and technical reliability, will guide ongoing improvements. Sustainable adoption will be supported through teacher training programs and student ambassador networks. The methodology will emphasize cultural relevance by integrating localized content examples in English and Chichewa.

TARGET AUDIENCE

1. Primary Audience: Students and Teachers

Learn Malawi will primarily serve Primary Schools Leaving Certificate(PSLC), Junior Certificate of Education (JCE) and Malawi School Certificate of Education (MSCE) students across the nation's diverse primary and secondary school system, including those in urban, rural, government, private, and faith-based institutions. It will also support both primary and secondary school teachers by providing comprehensive digital teaching resources, professional development materials, and classroom tools aligned with the Malawi national curriculum.

2. Secondary Audience: Education Policymakers and School Administrators

Education policymakers and administrators represent the secondary audience, including officials from the Ministry of Education, district education managers, head teachers, and examination board members. These stakeholders will leverage the platform's extensive data analytics capabilities to track curriculum implementation, monitor learning outcomes, and inform evidence-based decisions on resource allocation.

3. Tertiary Audience: Researchers, Development Partners, and Global Stakeholders

The tertiary audience consists of education researchers, development partners, and international organizations engaged in supporting Malawi's education system. This includes academic researchers analyzing learning outcomes, NGO staff assessing program impact, and global institutions such as UNESCO, UNICEF and the World Bank involved in education in

AI INTERGRATION

AI Tool Selected: Groq API with Llama 3.3

After evaluating several AI solutions against the constraints of the Learn Malawi project including cost, internet dependency, model quality, and alignment with the Malawian educational context the platform will utilize the Groq API, running Meta's Llama 3.3 70B large language model. Table 1 summarizes the evaluation criteria and outcomes.

Table 1: AI Tool Evaluation Matrix

Criterion	Groq + Llama 3.3	OpenAI GPT-4	Google Gemini	Local Ollama
Cost	Free (14,400 req/day)	Paid — ~\$0.03/1K tokens	Free tier available	Free but hardware-dependent
Credit card required	No	Yes	No	No
Response speed	Under 1 second	2–5 seconds	1–3 seconds	10–30 seconds
Offline capability	No	No	No	Yes (requires 16GB RAM)
Model quality	Excellent (70B parameters)	Excellent	Very good	Good (smaller models)
Malawi context awareness	Strong	Strong	Strong	Varies by model
Infrastructure needed	None, cloud API	None, cloud API	None, cloud API	Dedicated server with GPU

Groq's key differentiator is its proprietary Language Processing Unit (LPU) hardware, which delivers AI inference speeds significantly faster than conventional GPU-based cloud providers. For a student platform where engagement and patience are critical factors, sub-second AI responses meaningfully improve the user experience compared to the 3–5 second latency of competing services (Groq, 2024). At 14,400 free API requests per day, the platform can serve approximately 1,440 students each asking 10 questions daily at no cost more than sufficient for the platform's current scale and early growth phase.

The Llama 3.3 70B model, developed by Meta AI and made available under an open license, has demonstrated strong multilingual capabilities and performs well on educational reasoning tasks. Critically, the model has been observed to incorporate culturally relevant Malawian references including local place names, Chichewa greetings, and contextually appropriate examples without explicit instruction, making it well-suited for the Learn Malawi context.

3. Integration Architecture

The AI layer is embedded within the existing Learn Malawi technical stack without introducing new infrastructure dependencies. The platform's NestJS REST API (backend) hosts a dedicated AI Module that proxies' requests from the React frontend to the Groq API. This server-side architecture is critical for two reasons: it keeps the Groq API key secure on the server, and it allows the platform to implement rate limiting, caching, and logging without exposing these controls to the client.

Table 2: AI Integration Architecture Summary

Layer	Component	Role
Presentation	React 19 / Vite AI Chat Widget	Student-facing interface; sends user questions to the NestJS API
Application	NestJS AI Module /ai/ask endpoint	Applies context-specific system prompts; proxies to Groq; returns AI reply
AI Inference	Groq API Llama 3.3 70B	Generates natural language responses tuned to Malawian curriculum and student level
Data	PostgreSQL quiz attempts, performance logs tables	Stores student performance data used to generate adaptive quiz recommendations

Each AI request carries a feature identifier (e.g., tutor, quiz explain, career advice) that determines which system prompt is applied before the student's message is sent to Groq. This prompt engineering approach ensures the AI responds appropriately to the specific educational context a quiz explanation response is structured differently from a career guidance response, even though both route through the same endpoint.

4. AI-Powered Features

The AI integration introduces five distinct features, each mapped to an existing platform section. Together they transform the platform from a static resource library into an adaptive learning environment.

4.1 AI Study Tutor

Available on every page of the platform, the AI Study Tutor is a floating chat widget that students can open at any time to ask questions about any subject in the Malawi national curriculum. The tutor is configured to respond at the appropriate academic level JCE or MSCE and to use examples drawn from the Malawian context wherever possible.

Interaction example: A Form 3 student reading study notes on quadratic equations opens the tutor and types "I do not understand how to factorize." The AI responds with a step-by-step explanation using simple numbers, followed by a practice problem tailored to the student's query. This replicates one-on-one teacher support, which is scarce in most school environments where teacher-to-student ratios can exceed 1:80 (MOE, 2023).

Key benefits: 24/7 availability, instant response, no stigma of asking a "simple" question in front of peers, culturally relevant examples.

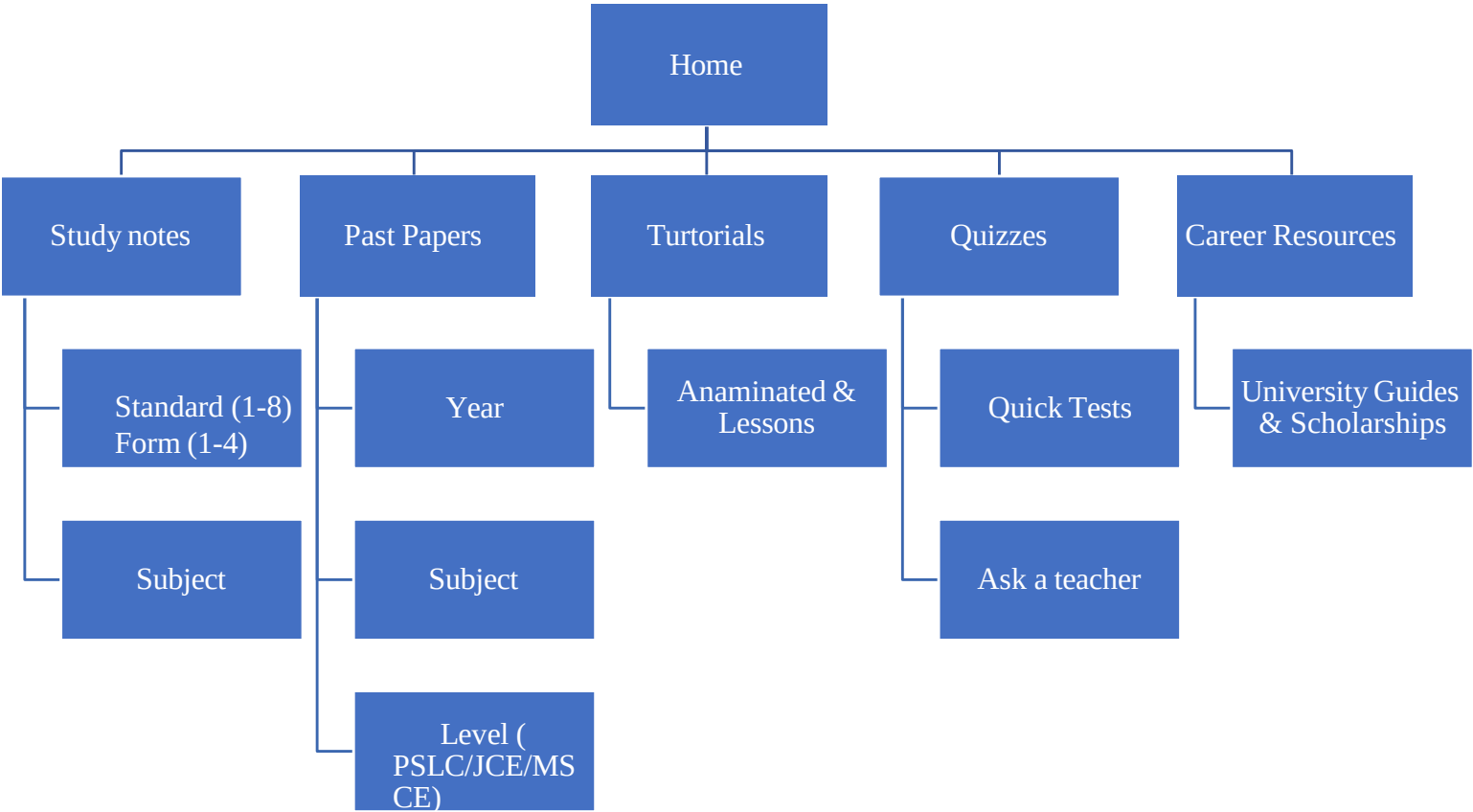
4.2 Adaptive Quiz Generation and Performance Analysis

This is the most pedagogically significant AI feature. After a student completes a quiz, the platform records their performance at the question level not just the overall score. The AI analyses the pattern of incorrect answers to identify specific knowledge gaps and then generates a personalized follow-up quiz targeting those exact weaknesses.

For example, a student who completes a Form 4 Mathematics quiz and scores 60% overall but answers all probability questions incorrectly will receive an AI-generated follow-up quiz focused exclusively on probability, with questions at a slightly lower difficulty level to rebuild foundational understanding before progressing.

This approach is grounded in Bloom's (1984) mastery learning theory, which demonstrates that students who receive targeted feedback on specific gaps achieve significantly higher outcomes than those who receive generalized instruction. The AI operationalizes this theory at scale, providing individualized mastery pathways to every student on the platform .

BLUEPRINT



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